

E3 Trigonometric functions

Name: _____

Class: _____

Date: _____

Time: **145 minutes**

Marks: **121 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

State the interval for which $\sin x$ is a decreasing function for $0^\circ \leq x \leq 360^\circ$

(Total 2 marks)

Q2.

- (a) Determine a sequence of transformations which maps the graph of $y = \cos \theta$ onto the graph of $y = 3\cos \theta + 3\sin \theta$

Fully justify your answer.

(6)

- (b) Hence or otherwise find the least value and greatest value of

$$4 + (3\cos \theta + 3\sin \theta)^2$$

Fully justify your answer.

(3)

(Total 9 marks)

Q3.

- (a) Write down the two solutions of the equation $\tan(x + 30^\circ) = \tan 79^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$.

(2)

- (b) Describe a single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan(x + 30^\circ)$.

(2)

- (c) (i) Given that $5 + \sin^2 \theta = (5 + 3 \cos \theta) \cos \theta$, show that $\cos \theta = \frac{3}{4}$.

(5)

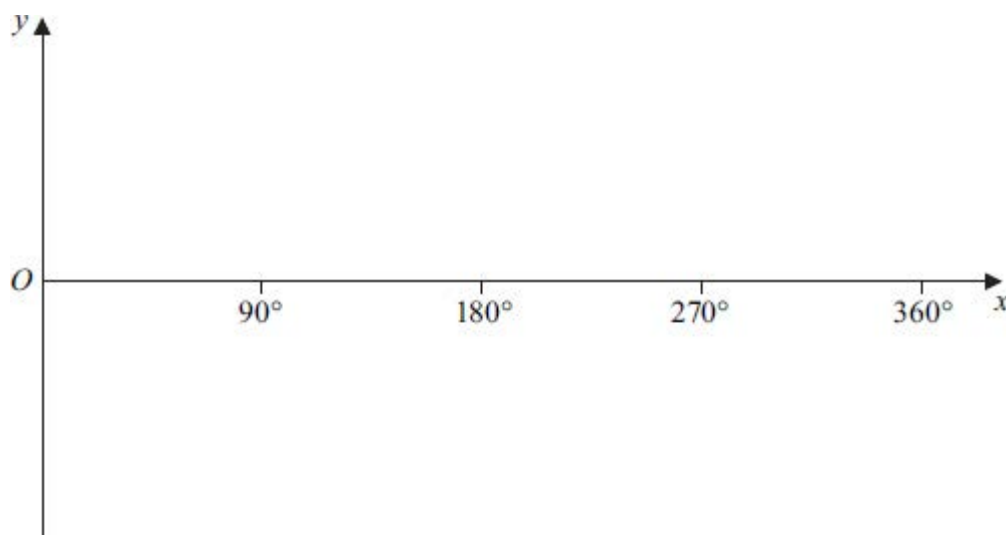
- (ii) **Hence** solve the equation $5 + \sin^2 2x = (5 + 3 \cos 2x) \cos 2x$ in the interval $0 < x < 2\pi$, giving your values of x in radians to three significant figures.

(3)

(Total 12 marks)

Q4.

- (a) (i) On the axes given below, sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$.



(3)

- (ii) Solve the equation $\tan x = -1$, giving all values of x in the interval $0^\circ \leq x \leq 360^\circ$.

(2)

- (b) (i) Given that $6 \tan \theta \sin \theta = 5$, show that $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$.

(3)

- (ii) **Hence** solve the equation $6 \tan 3x \sin 3x = 5$, giving all values of x to the nearest degree in the interval $0^\circ \leq x \leq 180^\circ$.

(6)

(Total 14 marks)

Q5.

The acute angles α and β are given by $\tan \alpha = \frac{2}{\sqrt{5}}$ and $\tan \beta = \frac{1}{2}$.

- (a) (i) Show that $\sin \alpha = \frac{2}{3}$, and find the exact value of $\cos \alpha$.

(2)

- (ii) Hence find the exact value of $\sin 2\alpha$.

(2)

- (b) Show that the exact value of $\cos(\alpha - \beta)$ can be expressed as $\frac{2}{15}(k + \sqrt{5})$, where k is an integer.

(4)

(Total 8 marks)

Q6.

Angle α is acute and $\cos \alpha = \frac{3}{5}$. Angle β is **obtuse** and $\sin \beta = \frac{1}{2}$.

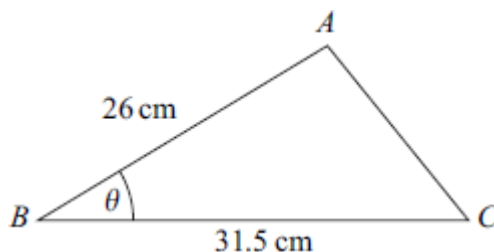
(a) (i) Find the value of $\tan \alpha$ as a fraction. (1)

(ii) Find the value of $\tan \beta$ in surd form. (2)

(b) Hence show that $\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$, where m and n are integers. (3)
(Total 6 marks)

Q7.

The triangle ABC , shown in the diagram, is such that $AB = 26$ cm and $BC = 31.5$ cm.



The acute angle ABC is θ , where $\sin \theta = \frac{5}{13}$.

(a) Calculate the area of triangle ABC . (2)

(b) Find the exact value of $\cos \theta$. (1)

(c) Calculate the length of AC . (3)
(Total 6 marks)

Q8.

(a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$. (2)

(ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$. (2)

(b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ . (3)

- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P .

(3)

(Total 10 marks)

Q9.

- (a) Sketch the graph of $y = \cos x$ in the interval $0 \leq x \leq 2\pi$. State the values of the intercepts with the coordinate axes. (2)

- (b) (i) Given that

$$\sin^2 \theta = \cos \theta (2 - \cos \theta)$$

prove that $\cos \theta = \frac{1}{2}$.

(2)

- (ii) Hence solve the equation

$$\sin^2 2x = \cos 2x (2 - \cos 2x)$$

in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

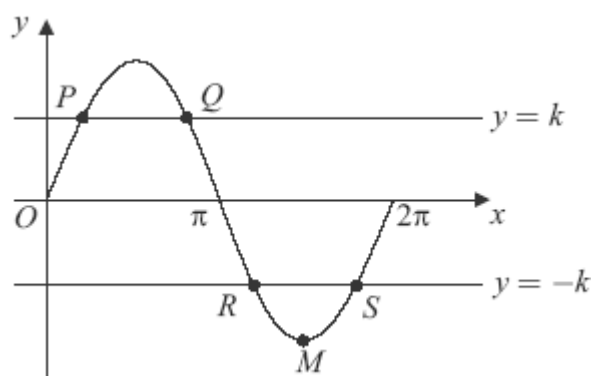
(4)

(Total 8 marks)

Q10.

- (a) Solve the equation $\sin x = 0.8$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures. (3)

- (b) The diagram shows the graph of the curve $y = \sin x$, $0 \leq x \leq 2\pi$ and the lines $y = k$ and $y = -k$.



The line $y = k$ intersects the curve at the points P and Q , and the line $y = -k$ intersects the curve at the points R and S .

The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of the point Q in terms of π and α .

(1)

- (iii) Find the length of RS in terms of π and α , giving your answer in its simplest form.

(2)

- (c) Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$, indicating the coordinates of points where the graph intersects the x -axis and the coordinates of any maximum points.

(5)

(Total 13 marks)

Q11.

- (a) The angle α is acute and $\sin \alpha = \frac{4}{5}$.

- (i) Find the value of $\cos \alpha$.

(1)

- (ii) Express $\cos(\alpha - \beta)$ in terms of $\sin \beta$ and $\cos \beta$.

(2)

- (iii) Given also that the angle β is acute and $\cos \beta = \frac{5}{13}$, find the exact value of $\cos(\alpha - \beta)$.

(2)

- (b) (i) Given that $\tan 2x = 1$, show that $\tan^2 x + 2 \tan x - 1 = 0$.

(2)

- (ii) Hence, given that $\tan 45^\circ = 1$, show that $\tan 22 \frac{1}{2}^\circ = \sqrt{2} - 1$

(3)

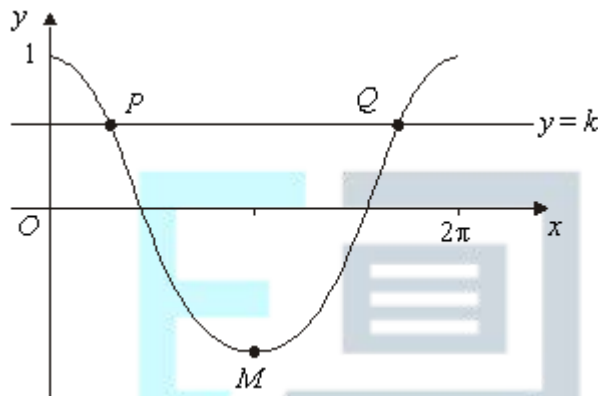
(Total 10 marks)

Q12.

- (a) Solve the equation $\cos x = 0.3$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures.

(3)

- (b) The diagram shows the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$, and the line $y = k$.



The line $y = k$ intersects the curve $y = \cos x$, $0 \leq x \leq 2\pi$, at the points P and Q .
The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of Q in terms of π and α .

(1)

- (c) Describe the geometrical transformation that maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$.

(2)

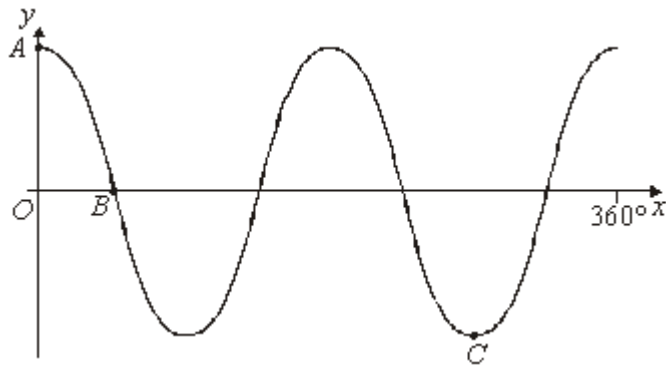
- (d) Solve the equation $\cos 2x = \cos \frac{4\pi}{5}$ in the interval $0 \leq x \leq 2\pi$, giving the values of x in terms of π .

(4)

(Total 12 marks)

Q13.

The diagram shows the graph of $y = \cos 2x$ for $0^\circ \leq x \leq 360^\circ$.



- (a) Write down the coordinates of the points A , B and C marked on the diagram. (4)
- (b) Describe the single geometrical transformation by which the curve with equation $y = \cos 2x$ can be obtained from the curve with equation $y = \cos x$. (2)
- (c) Solve the equation

$$\cos 2x = 0.37$$

giving all solutions to the nearest 0.1° in the interval $0^\circ \leq x \leq 360^\circ$.
 (No credit will be given for simply reading values from a graph.)

(5)
(Total 11 marks)